XENONnT Ionization Data Constrain Solar Chameleons

First sub-keV search for solar chameleons using S2-only data sets new limits.

arXiv:2609.33198 — XENONnT ionization-only search for solar chameleons.

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Background

Screened scalar fields link dark energy to laboratory-scale new physics while evading fifth-force constraints. Solar production offers a direct probe, and ionization-only measurements extend liquid-xenon sensitivity into the sub-keV regime.

> **No prior search for solar chameleons using sub-keV ionization-only data existed.**

What this paper is selling

This analysis establishes low-energy ionization data as a new channel for testing solar chameleons. It folds the predicted electronic-recoil spectrum through run-dependent detector response and fits seven corrected-S2 bins across three science runs.

Key results to remember

- First solar chameleon search with XENONnT ionization-only data reaches 0.04 keVee.
- Disformal-dominated limit: log10 βeff < −3.64 at 95% credible level.
- Weaker than previous electronic-recoil limit due to falling signal and rising background.

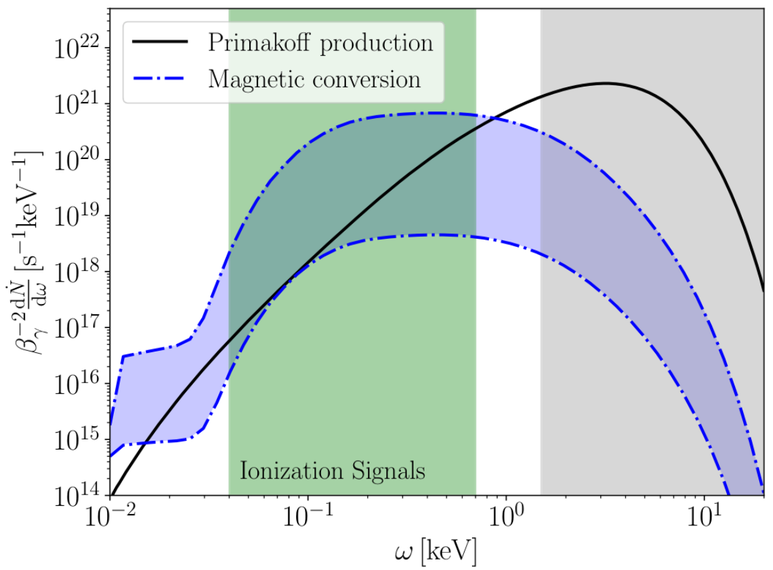


Figure 1 | Solar chameleon production spectrum shows Primakoff and magnetic conversion contributions, with the green band marking the XENONnT analysis window.

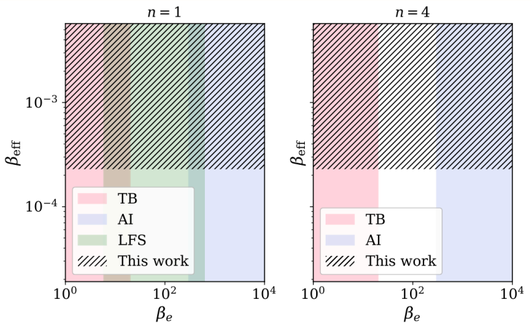


Figure 6 | Excluded regions in the βeff–βe plane for n=1 and n=4, compared with atom interferometry, levitated force sensors, and torsion-balance bounds.